CHIH-MEI YOUNG

+1 7345100952 / ychihmei@umich.edu

EDUCATION

University of Michigan, Ann Arbor, MI, USA **Sep 2021 – present**
Doctor of Philosophy in Chemical Engineering & Scientific Computing (GPA: 3.9/4)
- advisor: Prof. Michael J. Solomon & Prof. Sharon C. Glotzer

National Tsing Hua University (NTHU), Hsinchu, Taiwan **Sep 2018 – Jun 2020**
Master of Science in Chemical Engineering (GPA: 4.12/4.3)
- advisor: Prof. Hsin-Lung Chen

National Tsing Hua University (NTHU), Hsinchu, Taiwan **Sep 2014 – Jun 2018**
Bachelor of Science in Chemical Engineering (GPA: 3.91/4.3)

RESEARCH EXPERIENCES

Ph.D. Degree Thesis: ***Functional Properties of Self-assembled Colloids by Experiment and Simulation***

- Integrated experimental techniques with molecular dynamics (MD) simulations to investigate external field–driven colloidal crystal formation and propulsion dynamics of anisotropic active colloids
- Performed MD simulations of active nanorods within inverse opal structures to investigate their transport dynamics and confinement effects (in collaboration with Prof. Daniel K. Schwartz in University of Colorado, Boulder)
- Engineered spherical and ellipsoidal active particles with platinum coatings via physical vapor deposition (PVD) in a cleanroom environment and characterized particle configurations with scanning electronic microscopy (SEM)
- Developed machine learning frameworks for high-throughput trajectory analysis to identify active patch geometry, orientation, and dynamic behavior of active particles

Master's Degree Thesis: ***Columnar Mesophases of Dendrimer-Surfactant Complex***

- Employed small and wide-angle X-ray scattering technique (S/WAXS) to probe the nanostructures of the complexes
- Provided controllable structure transitions of dendrimer-surfactant complexes for further applications

Process Design Project: ***Design of a Chemical Factory Yielding 100 Thousand Ton of Cumene per Year***

- Utilized ASPEN plus to design the manufacturing process and evaluated the cost-effectiveness of the production

WORKING EXPERIENCE

Department of Chemical Engineering, NTHU, Hsinchu, Taiwan **Jul 2020 – Aug 2021**
Research Assistant, Prof. Hsin-Lung Chen's Polymer Physics Lab

- Studied the effect of the dispersion state of nucleating agents on the crystallization kinetics and mechanical properties of poly(butylene terephthalate) (PBT) produced by Chang Chun Plastic Group, Taiwan
- Optimized PBT/nucleating agent system to enhance the crystallization rate while largely retain polymer ductility
- Conducted rheological experiments to evaluate the ductility and mechanical behavior of PBT

Department of Chemical Engineering, University of Michigan, Ann Arbor, MI **Jan 2025 – May 2025**
Teaching Assistant of Undergraduate Unit Operation Lab Course

- Guided undergraduate students through the unit operation experiments and provided instruction on data analysis

United Microelectronics Corporation, Singapore Branch (UMC SG) **Jul 2018 – Aug 2018**
Summer Intern, Department of Special Process Integration (SPI)

- Contributed to optimizing the electrical properties and production costs by modifying the manufacturing process
- Acquired advanced knowledge of semiconductor devices and gained hands-on experience in fabrication processes

TECHINICAL SKILLS

- Experimental techniques: WAXS, SAXS, XRD, SEM (EDAX), Confocal Laser Scanning Microscopy, DSC, TGA, NMR, FTIR, UV-Vis, Rheometer, Semiconductor processing experience (cleanroom environment, PVD)
- Programming languages: Python, C++, MATLAB (Simulink), SQL
- MD Simulation: HOOMD-blue, LAMMPS, GROMACS
- Computer-aided design/engineering: ANSYS (FDTD), COMSOL, Moldex3D, ASPEN Plus, Solidworks, Arduino
- High-performance computing (HPC): Parallel computing on Linux clusters, job scheduling and scripting
- Machine learning and image segmentation tools: CNN, YOLOv8, CVAT

PUBLICATIONS & APPEARANCES

1. Young, C.-M.; Moore, T. C.; Glotzer, S. C.; Solomon, M. J., Effect of Janus Balance and Shape Anisotropy on the Propulsion Dynamics of Active Colloidal Particles. AIChE Annual Meeting **2025** (oral presentation)
2. Liu, T.; Young, C.-M.; Moore, T. C.; Glotzer, S. C.; Solomon, M. J., Defect Structures in Colloidal Crystals and Their Effect on Grating Diffraction Structural Color. *ACS Applied Materials & Interface* **2025**
3. Mansel, B.W.; Su, C.-J.; Chen, C.-Y.; Young, C.-M.; Huang, Y.-C.; Yang, C.-C.; Chen, H.-L., Superhelical DNA liquid crystals from dendrimer-induced DNA compaction. *Soft Matter* **2021**
4. Young, C.-M.; Chang, C.-L.; Chen, Y.-H.; Chen, C.-Y.; Chang, Y.-F.; Chen, H.-L., Dendrimer-mediated Columnar Mesophase of Surfactant. *Soft Matter* **2020**
5. Young, C.-M.; Chang, Y.-F.; Chen, Y.-H.; Chen, C.-Y.; Chen, H.-L., Ribbon Phase of Dendrimer–Surfactant Complexes. *Macromolecules* **2019**
6. Young, C.-M.; Chen, H.-L., Columnar Mesophases of Dendrimer-Surfactant Complex. Presented at Pacific Polymer Conference, Singapore, **Dec 2019** (oral presentation)

AWARDS & SCHOLARSHIPS

Third Place, University of Michigan Chemical Engineering Symposium Poster Competition, 2023
- Awarded 3rd place among 50+ participants for research poster presentation

Study Abroad Scholarship, Ministry of Education (Taiwan), 2022
- Top 10% of Taiwanese students received $32,000 USD for foreign education

Rackham Research Grant, University of Michigan, 2021
- Received a $1,500 research grant to support graduate research activities

Poster Competition, Taiwan Institute of Chemical Engineers Annual Meeting (TwIChE), 2020
- Received honorable mention with the topic "Dendrimer-mediated Columnar Mesophase of Surfactant"

LCY Education Foundation 9th Scholarship, 2019
- Received NT$40,000 scholarship (approximately $1282 USD)

Process Design Contest, TwIChE, 2018
- Awarded honorable mention among 30+ teams applying ASPEN plus to chemical engineering process simulation and design

EXTRACURRICULAR ACTIVITIES

Outreach Activity Organizer, University of Michigan Summer Engineering Exploration Camp, 2025
- Led the "Crystal Crafts" activity for 40 high school students to spark interest in STEM
- Engaged participants in hands-on learning by guiding them to self-assemble colloidal particles into colorful crystal keychains, connecting science with creativity

Finalist, University of Michigan Engineering Symposium Three Minutes Thesis (3MT), 2023 & 2025
- Delivered a clear, thee-minute presentation distilling complex thesis research for a broad audience
- Selected as one of 20 finalists from a pool of 60+ competitive engineering students

ChE Symposium Chair, University of Michigan Chemical Engineering Annual Symposium, 2023
- Organized a department-wide symposium with 150+ attendees
- Secured a keynote speaker and engaged industry participants for sponsorship opportunities

Team Experiment & Product Designer, NTHU Innovative Competition – Microfluidic Device, 2018
Team Project Manager, Beijing-Taiwan Startup Contest – Microfluidic Device, 2018
- Developed an innovative remedy for ring-peeling of woody plants using microfluidic devices
- Demonstrated the prototype of the product and the proposal of business model
- Orchestrated the development progress and design of the product and won awards in both contests

Wet Lab Team Member, International Genetically Engineered Machine Competition (iGEM), 2016
- Found a method to degrade PFC (perfluorinated compounds, a category of pollutants) with genetically modified *E.coli* and won "Bronze Medal" in iGEM Giant Jamboree, 2016
- Conducted biological experiments, such as PCR, cloning, bacteria cultivation, SDS-page, etc.
- Communicated and cooperated with team members majoring in diverse subjects

REFERENCE

Michael J. Solomon, Professor
Department of Chemical Engineering, University of Michigan
mjsolo@umich.edu / (734) 764-3119

- Relationship: PhD Thesis Advisor

Sharon C. Glotzer, Professor
Department of Chemical Engineering, University of Michigan
sglotzerkjc@umich.edu / (734) 615-6296

- Relationship: PhD Thesis Advisor

Daniel K. Schwartz, Professor
Department of Chemical & Biological Engineering, University of Colorado, Boulder
daniel.schwartz@colorado.edu / (303) 735-0240

- Relationship: Research Project Collaborator